

FAIR Enough? Assessing Open Science Practices on a Top-Tier Security & Privacy Conference

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Abstract—In this paper, we investigate the practices surrounding the publication of research artefacts in the Security and Privacy research domain through the lens of Open Science and the FAIR (Findable, Accessible, Interoperable, and Reusable) principles.

We present an exploratory study of papers published at the IEEE S&P Symposium 2024 to assess voluntary artefact sharing in the absence of mandatory disclosure policies. While many researchers demonstrate a commitment to openness, our analysis reveals a recurring gap between the intent to share and the actual availability of functional pointers to their artefacts within the text. In addition to simply not providing access to artefacts, we identify several systemic barriers to accessibility, ranging from false claims to avoidable clerical errors.

Our findings suggest that current, non-standardized practices often lead to ambiguities that undermine the findability and reusability of research components. To address these challenges, we advocate for Open Science by promoting not only the sharing of research artefacts aligned with the FAIR principles but also the verifiability of research and its contribution to cumulative scientific advancement.

1. Introduction

Security and Privacy (S&P) within the digital space are growing research fields. As a subfield of Computer Science (CS), it makes many novel contributions that produce artefacts such as programs or source code. These can range from programs for analysing a phenomenon (e.g., a web crawler) to programs for detecting specific vulnerability classes (e.g., a fuzzer or a static analysis framework) to machine-checkable cryptographic proofs or data collected during web crawling experiments. In some cases, it may be necessary to reproduce the author’s work to validate their contributions. While authors attempt to provide sufficient details to reproduce their results, doing so in full is often not feasible in the space of a research paper. In such cases, to effectively reproduce a paper’s contribution, reviewers must be given access to the original artefacts used by the authors. This issue is known to conference chairs, and to facilitate replication, venues have begun to encourage authors

to participate in the artefact Evaluation (AE) procedure [30] or even have made it mandatory [7].

Another aspect to consider is the publication venues’ efforts towards Open Science (OS) in general, e.g., by providing open access to all publications [31]. Besides reproducibility, providing access to artefacts also fosters transparent research and cumulative progress, as other researchers can build on them [8, 1]. Some venues even enforce an “Open Science” statement within the final paper, including public access to the artefacts used [27, 10]. These approaches clearly demonstrate the trend towards implementing OS within the CS community.

In this paper, we examine the extent to which authors follow this trend beyond the publication venues’ mandatory guidelines. For this purpose, we conduct an exploratory study by examining publications from one top-tier venue in the domain of Privacy and Security in CS that, according to its *Call for Papers*, does not enforce access to the papers’ artefacts. Concretely, for papers that produced artefacts, we are interested in whether the authors provide pointers to them in the actually published texts, e.g., to publicly accessible repositories or databases. And where not, if they justify this. As we adhere to blame-free research, we will not make explicit references to papers that we consider *problematic* in the context of the discussed issue. However, as the papers we examine are publicly available, some examples may still be recognisable. Please note that it is not our intention to point fingers at others, but rather to reflect on our own experiences.

Contribution. We first introduce the underlying concepts of OS and the FAIR Principles, justifying the need for artefact publication. We then introduce the methodology used to examine papers published at the Security & Privacy Symposium 2024 [9] and evaluate the results collected. Finally, we discuss the various cases of artefact handling we observe in the published papers and reflect upon our insights in the conclusion.

Scope. It is not the intention of our work to evaluate research artefacts in terms of their suitability, quality, and long-term availability. This matter is of paramount importance,

but it concerns a different problem and is already under discussion in the research community [33, 14]. We further do not strive to conduct a quantitative, i.e., large-scale, study spanning multiple conferences and publication cycles. Instead, we focus on a single conference to observe how the OS approach, specifically artefact publications, is accepted and implemented by the S&P community without any enforcement mechanisms.

2. Background

There is a growing demand for a transparent research approach within the CS community. According to Elliot, “researchers can be transparent about many different elements of their practice, including their data, methods, computer codes, analysis plans, conflicts of interest, and value judgements” [13]. This view of transparency is closely linked to the notion of OS in science and research. In this section, we discuss the concept behind OS for scientific research and the related FAIR principles.

2.1. Open Science (OS)

OS aims to ensure the free availability and usability of scholarly publications, the data resulting from scholarly research, and the methodologies, including code or algorithms, used to generate those data [24]. In this section, we focus on the openness of research components, e.g., artefacts and datasets, and the extent to which this openness can be considered a necessary condition for good scientific practice.

In the following, we want to sketch two arguments for open research components: First, mutual verification of scientific knowledge enables reproducibility of research results and provides an opportunity for scientific quality control. Second, sharing knowledge enables the cumulative advancement of the field.

The communal character of science, the sharing and dissemination of knowledge, is considered as a cornerstone of the scientific ethos [22]. In the present day, this idea is represented by the Open Science Movement, which aims to foster the dissemination and reuse of research components, i.e., datasets, methods, and artefacts [21].

OS is consistent with other values of scientific research, such as the critical questioning of research outputs, i.e., assessing their robustness and validity. OS is a way to enable critical feedback without relying on opaque peer review mechanisms. The open access and traceability of research components are therefore a significant part of the scientific ethos and are desirable, regardless of the institutional effects of conferences, journals, and funding bodies. We consider the opening of one’s research components to scrutiny not just as another top-down regulatory measure, but as good scientific practice.

Nevertheless, there are cases where not being transparent in scientific research can be justified. Especially in fields that work with vulnerable groups, where the stakes of personal harms are high, commercial entities using open research components for commercial purposes (a common problem

in Open Source Software), and the distortion of scientific results in bad faith [21]. However, these caveats are not a valid reason to keep research closed. These cases must be justified individually, and the values of Open Science and harm prevention must be carefully weighed against one another.

This commitment to opening up science and scientific knowledge is incorporated into the principles and practices of conferences, journals, and research funding bodies. This includes public, trustworthy repositories, the obligation to publish open access when possible, and research data management plans.

2.2. FAIR Principles

As this paper focuses on technical components (e.g., artefacts, code, datasets), we will limit our discussion to research data management and the FAIR principles. These principles are already required by the European Commission’s European Research Council [26]. FAIR stands for *Findability*, *Accessibility*, *Interoperability*, and *Reusability* of digital resources and does not prescribe any specific implementation, but rather offers guidance on finding flexible solutions for these requirements [32].

- **Findable:** This principle describes practices for making digital resources findable and ensuring that they continue to be findable. This is implemented via specific mechanisms, such as issuing globally unique, persistent identifiers (e.g., project-agnostic identifiers) to ensure their longevity, using metadata to enable unambiguous identification of resources, and indexing them as machine-readable, searchable resources.
- **Accessible:** This principle describes the open accessibility of research components. This includes how the data is made accessible (e.g., via a trusted repository), the methods and tools needed to access the data, software documentation, and clear access management if restrictions apply (e.g., a mechanism to grant access to authorised users).
- **Interoperable:** This principle describes the allowance of data exchange and reuse between researchers, institutions, and organisations. Researchers should adhere to clear standards and, ideally, use open-source software to ensure interoperability.
- **Reusable:** This principle deals with licences and access restrictions for research components. The aim is to increase data re-use through clear licences and, if embargoes apply, to specify when the research data will be accessible and which measures will be used to ensure data quality.

Regardless of whether one considers openness to criticism and the verifiability of research to be a necessary part of good scientific practice (which we do), the standardisation of research data handling by institutions like the ERC should not be underestimated. The rule of thumb for ERC-funded projects is “as open as possible, as closed as necessary”. Generally, a blanket refusal to grant access to research materials is not an option if one wants access to these research funds.

3. Methodology

To conduct our exploratory study to investigate how authors adhere to artefact publication, we need to analyse each published paper to identify references (mentions) of the artefact provided with the publication.

Dataset sample. We select papers published in the Security & Privacy Symposium 2024, where the entire corpus comprises 244 papers. We have chosen this conference because it does not specify binding requirements for the provision and publication of artefacts in this cycle. At the same time, this publication cycle is not that far back, so its results can well reflect the current approaches to artefact publications.

Phrase Detection. Our strategy for detecting artefacts in papers relies on automatically identifying phrases associated with the description of an artefact being made publicly available in academic work. We manually review papers and identify 104 phrase variations commonly used to describe an artefact. In this phrase-detection process, we identify the paragraph that mentions the availability of an artefact and its corresponding pointer (e.g., a link in the text, a footnote, or a reference in the work's bibliography). The result consists of a table that maps each paper to the keywords found, the corresponding text extract, and pointers to any artefacts.

Validation Process. After the detection process, we manually validate the resulting papers, totaling 108. These papers include those in which we find no mention of artefacts, and those that mention artefacts but provide no pointers to them. This manual analysis is essential because, for example, papers may mention the availability of an artefact but provide no link within the paper. Furthermore, our tool may also return a false negative, where the phrase found does not refer to an artefact of the paper in question, but, e.g., to any related work.

Output Result. The outcome of this process of information extraction and manual validation generates a dataset that allows us to observe the practice of authors providing access to their research artefacts voluntarily, therefore adhering to the FAIR principles (Section 2.2). As mentioned in Section 1, we adhere to blame-free research. Therefore, during the peer review of our paper, we are committed to providing our evaluation data to the PC chairs upon request, but refrain from including them in the final publication. However, the source code and the phrases used in our study are publicly available¹.

4. Evaluation

In total, we examine a set of **244** papers, published at the *Security & Privacy Symposium 2024* [9]. Using the methodology introduced in Section 3, we first identify 57 papers from the entire corpus that presumably did not

generate any artefacts. We review those papers manually to avoid false negatives before excluding them from the subsequent evaluation.

The remaining set contains 187 papers that produced artefacts of any form. It is initially divided into two groups:

- 1) 156 papers that provide pointers to their artefacts, usually in the form of a publicly accessible URL placed anywhere in the text; and
- 2) 31 papers in which no pointers to their artefacts produced could be detected within the text.

For the first group, the majority of the papers provide a publicly accessible pointer to their artefacts, identified by a link directly within the text, a footnote, or a reference. Usually, those links point to a public repository on a platform such as GitHub. However, we need to note that we do not claim completeness at this point. Because our detection relies on manually curated phrases that indicate the publication of artefacts, our approach is susceptible to false positives. However, we consider this tolerable, as we are primarily interested in examining samples of negative cases.

The second group of papers (31), which produced any form of artefacts, has been manually reviewed by our authors, resulting in three categories:

- (a) 24 papers do not provide any publicly accessible pointers to their artefacts within the text, or discuss any plans to do so;
- (b) 2 papers justify why the authors refrain from publishing their artefacts. In both cases, ethical considerations are the reason, as the artefacts contain sensitive user data. In one case, the authors promise to provide access to the data upon request via direct contact; and
- (c) 6 papers promise that their artefacts would be publicly available, but fail to provide a valid pointer in the publication's text for accessing them.

The reasons those 6 papers fail to fulfil their open-source claims vary. In one case, the authors fail to provide a correctly rendered clickable hyperlink text, resulting in: "[...] we published our code at LINK." In another case, the authors seem to have forgotten to update the camera-ready version of their paper: "<link is disclosed upon publication>". In the remaining cases, the authors simply claim that their artefacts are publicly available. However, a pointer to them is nowhere to be found in the corresponding paper.

5. Discussion

The evaluation of papers published in a high-ranked S&P conference provides insights into how the research community adapts OS approaches, namely, by providing public access to their artefacts without being forced to do so. In this Section, we discuss individual cases we observe in our data and map them to the FAIR principles, introduced in Section 2.2.

No Pointers to artefacts. We find that 13 % of the corpus's papers do not provide public access to their research artefacts in the text, which we consider non-negligible. Recall that

1. <https://doi.org/10.5281/zenodo.18698730>

due to our manually curated phrases, we expect a reasonable number of false positives. We have deliberately refrained from manually searching the Internet for any traces of artefacts in those papers, as we expect them to be linked where they are introduced – in the paper’s text. Furthermore, it remains unclear what one should search for: the artefact’s name, the paper’s title, or the author’s website? And even if the artefacts can be found through one’s own efforts, the repository description may not provide the detailed information needed to associate them with the original paper. Hence, this approach contradicts the principle of *findability*.

Use of Open Source Software (OSS). According to the methodologies described in papers, many authors utilise OSS for their research. In some cases, authors even extend or improve some OSS to support their own research objectives. This aligns with the discussion by Barba, indicating the essential role of OSS for research in general and reproducibility of its artefacts [4]. Fortunato and Galassi discuss the interface between free OSS and reproducibility and further connect this to research quality and sustainability [16]. However, many of the papers we examine have not made their resulting software publicly available. Given the rising demand for OS, this approach is at least questionable and contradicts the principle of *interoperability*.

Open-Sourcing artefacts. The vast majority of the published artefacts are hosted on the widely established platform github.com. Considering the principle of *accessibility*, it is debatable whether the major research community should rely on one single proprietary platform for hosting research artefacts. Apart from that, it must be acknowledged that open-sourcing their research artefacts is not a trivial task, as it is not simply a matter of uploading the data anywhere. Authors need to document their material in a way that allows researchers to understand the content and, if possible, reproduce it. And meeting the demand for reproducibility can be quite challenging: Some setups may require very specialised or proprietary hardware, or the resulting data may capture only a snapshot of a non-deterministic system (often the case with web measurement studies). In some cases, authors need to consider ethical implications, licensing and authorship policies, which may directly impact the process.

Provision Upon Request. Some authors promise to provide access to their research artefacts upon request via mail. However, sharing content with such restrictions needs careful consideration. An individual author may change their affiliation, resulting, for example, in the loss of access to their mail account. Furthermore, the restriction on artefacts allows authors to protect resources that may be harmed by malicious entities, e.g., by disclosing sensitive user data. Providing access upon request shifts this responsibility to other, possibly unknown entities, rendering the restriction’s protective rationale ineffective. To adhere to the *accessibility* principle, authors should exercise caution when restricting access to a resource and consider how this artefact will be maintained in the long term.

Lack of Standardisations. The pointers to artefacts are specified in various ways and placed in different parts within the paper’s text. Mostly, they are placed in the text flow, as a footnote, or in the reference section, and then cited in the text. The lack of uniform presentation makes it difficult to reliably identify such pointers. For the community to benefit from such artefacts, they need to be comprehensible, where possible reproducible in practice, and easily accessible. This lack of standardisation creates unnecessary barriers to access. The lack of standardisation on when, where, and in what form research artefacts should be published creates access limitations, ultimately impacting the long-term development of the process (*interoperability* principle). However, until today, the requirements for handling artefacts diverge widely across top-tier conferences [2, 3, 6].

Fostering Future Research. We observe papers advocating future scientific progress by introducing innovative tools, frameworks, and datasets intended to support the community. However, when these artefacts are not made openly available, the actual impact of such contributions becomes questionable. Without access, other researchers cannot verify the functionality, build upon their features, or adapt them to new research contexts. This undermines the principles of reproducibility and openness, which we consider fundamental to scientific advancement, adhering to the principle of *reusability*. Furthermore, enabling access to artefacts is essential for fostering long-term research in the community.

Institutionalisation. Institutionalising scientific practices (e.g., peer review, grant applications) provides cultural schemata that make these practices the norm, thereby avoiding subversion and rejection. The increasing institutionalisation of Open Science and the FAIR principles at conferences can help make these scientific practices the norm and a shared concept of good scientific practice within the broader S&P community [23]. A proposal to institutionalise other stages of the research procedure in CS, e.g., ethics review, has already been discussed [12]. At the same time, this approach raises questions about the trustworthiness of the state-of-the-art tools we currently use, for example, for hosting such artefacts. One option could be the establishment of AE tools maintained from within the community [14]. Another possibility is to allow specification of artefact pointers within the submission tools, enabling automated requirements checking and inclusion in the camera-ready version of the paper.

Hosting and Infrastructure Platforms. According to our data, GitHub is the preferred hosting platform for artefacts within the research community. However, we consider it the researchers’ responsibility to use suitable platforms to make artefacts available. The preferred platform may change depending on the research geolocation or institutional policies. Also, platform changes (e.g., changes to terms and conditions) may limit or restrict access to an artefact in the long term. We understand that a perfect independent hosting

infrastructure does not exist (yet) and may require significant financial and administrative resources [15]. However, this would be a useful initiative for academic freedom, long-term availability, and FAIR principles.

6. Related Work

Klemmer et al. [20] analysed 200 papers in the privacy and security community from 12 conferences. Initially, 52 transparency criteria were defined based on their paper investigation. With these criteria, it was found that while many papers meet some transparency criteria, none achieve full transparency in all 52 criteria, with essential materials often missing, and papers often differ in how and what transparency criteria are reported. They also recommend formalising guidelines and adapting artefact evaluation to improve reporting standards.

Olszewski et al. [25] measured the reproducibility of nearly 750 papers, their codebases, and datasets. They observed that over 60% of papers from top security conferences (2013-2022) did not share code, and only 43% of shared artefacts produced comparable results. Identifying major obstacles, including missing dependencies and poor documentation.

We can also mention research in the area that focuses on describing lessons learned from conducting the artefact review process [28, 11, 17], and the improvement of artefact reproducibility and reusability [34, 18, 5].

7. Conclusion

In this paper, we emphasise that openness extends beyond publications to encompass access to the artefacts produced in research projects. Our goal is to encourage the community not only to share artefacts meaningfully but also to do so in a manner that aligns with the FAIR Principles.

We are undoubtedly not the first to make this argument [19, 23]. However, the ambiguity of recent *Calls for Papers* for top-tier conferences, as well as our examination of published papers from S&P'24, highlight the ongoing inconsistency in how access to research artefacts is provided (or not) in CS papers.

Our exploratory study reveals that while a majority of authors voluntarily share their work, many artefact-producing papers fail to provide any direct pointers within the text. These findings highlight a critical gap in current practices, including technical oversights like broken hyperlinks and placeholder text, which frequently undermine the findability and accessibility of research artefacts.

Ultimately, we call on the S&P research community – including conference organisers and program chairs – to join forces to institutionalise Open Science as a shared concept of good scientific practice and to collaborate on standardising publication practices for artefacts. This collective effort is essential to ensure that research components are not merely shared, but are truly **F**indable, **A**ccessible, **I**nteroperable, and **R**eusable. By expanding our efforts towards clearer, uniform

standards, we can ensure that every contribution remains transparent and verifiable, fostering the cumulative progress and innovation necessary for the long-term advancement of S&P research.

Ethics considerations

This work does not raise any ethical issues, as we used only publicly available data and did not interact with the corresponding authors [29].

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